



GitHub Copilot Dev Days

Visual Studio 2026에서
GitHub Copilot 사용해보기



이종인

Microsoft MVP / DesignMe AI

저는요!

이종인

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- Microsoft MVP | Developer Technologies
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Visual Studio + GitHub Copilot



함께하는 AI



Visual Studio 2026

기능

-  디버거
-  프로파일러
-  테스트
-  현대화
-  GitHub

도구

- 계획
- Microsoft Learn
- NuGet
- GitHub
- Azure MCP 서버

& more

계획



Create Todo Console Application

Overview: Create Todo Console Application

Progress: 28%

Last Updated: 2025-11-06 19:16:01

Plan Steps

- ✓ Create Todo model class
- ✓ Create TodoService for business logic
- ✗ Create TodoRepository for data persistence

GitHub Copilot Chat

Create and implement a plan to create a todo app

Create and implement a plan to create a todo app
> 1 reference

GitHub Copilot
> 1 reference

I'll create a plan to build a todo app for your .NET 10 console application.

Plan Created [Open](#)

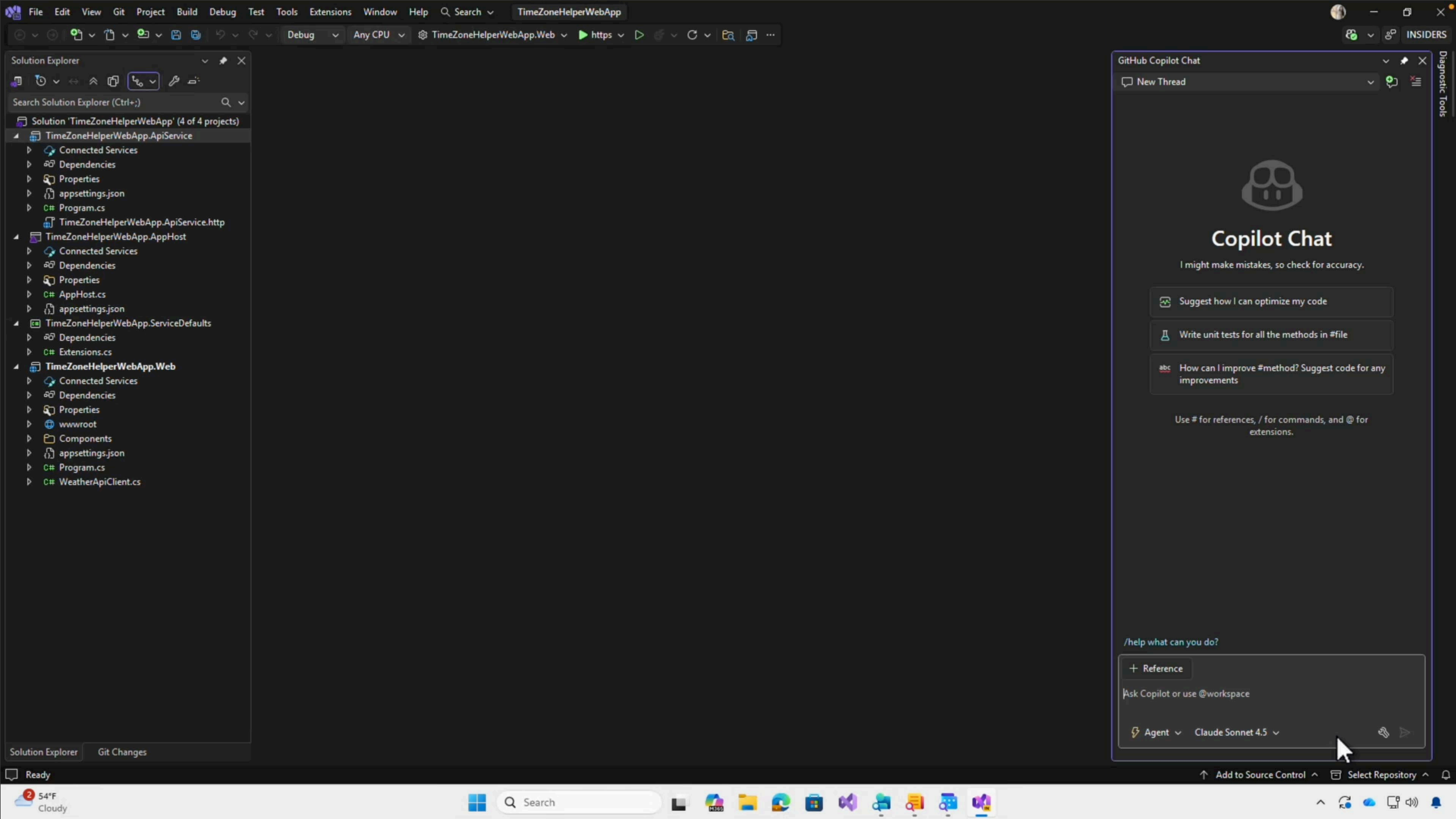
Now I'll start implementing the todo app. Let me begin with **Step 1: Create Todo model class.**

C# Todo.cs	ConsoleApp3\Models	-0	+15
------------	--------------------	----	-----

Now working on **Step 2: Create TodoService for business logic.**

C# TodoService.cs	ConsoleApp3\Services	-0	+82
-------------------	----------------------	----	-----

Now working on **Step 3: Create TodoRepository for data persistence.**



인라인 제안+ 에이전트모드 채팅

빌드



```
38  
39  
40  
41  
42  
43  
44 →  
45
```

```
1 reference | 0 changes | 0 authors, 0 changes  
public bool IsWithinWorkingHours(DateTime dateTime)  
{  
    var hour = dateTime.Hour;  
  
    // return times between 9 AM to 5 PM  
    return hour >= 9 && hour < 17;  
}
```

TimeZoneService.cs

```
74  
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111  
112  
113  
114  
115
```

```
return _timeZones.ToList();  
  
public bool AddTimeZone(string name, string id)  
{  
    if (string.IsNullOrEmpty(name) || string.IsNullOrEmpty(id))  
    {  
        return false;  
    }  
  
    // Check if time zone already exists  
    if (_timeZones.Any(tz => tz.Id == id))  
    {  
        return false;  
    }  
  
    // Validate that the time zone ID is valid  
    try  
    {  
        TimeZoneInfo.FindSystemTimeZoneById(id);  
    }  
    catch (TimeZoneNotFoundException)  
    {  
        return false;  
    }  
  
    _timeZones.Add(new TimeZoneEntry { Name = name, Id = id });  
    return true;  
}  
  
public bool RemoveTimeZone(string id)  
{  
    var timeZone = _timeZones.FirstOrDefault(tz => tz.Id == id);  
    if (timeZone != null)  
    {  
        _timeZones.Remove(timeZone);  
        return true;  
    }  
    return false;  
}
```

GitHub Copilot Chat

Add the ability for users to add additional time zones to t...

- Remove button styling
- Header actions layout
- Responsive design updates

Features:

- ☒ Users can add custom time zones with their own display names
- ☒ Users can select from all available system time zones
- ☒ Users can remove any time zone from the list
- ☒ Validation prevents duplicate time zones and invalid entries
- ☒ User-friendly feedback messages
- ☒ Clean, modern UI that matches the existing design
- ☒ Responsive design for mobile devices

The application is now ready to use with full add/remove functionality for time zones!

☒ Read TimeZoneService.cs

☒ Read Home.razor

✓ Total changes (3) Keep Undo ...

C# TimeZoneService.cs

Home.razor

Home.razor.css

+ Reference

Ask Copilot or use @workspace

Agent Claude Sonnet 4.5

100 % No issues found Keep Undo Ln: 70, Ch: 6 SPC CRLI

테스트



GitHub Copilot Chat

@test #C:\Users\cathys\source\repos\eShop-tes...

Test (Preview)

> 0 references

☒ Summary of Test Generation

Status: ☒ Completed Successfully

Total Unit Tests Generated: 59 tests

Files Created/Modified: 7 files

Test Projects Touched: 1 project

- Projects: Basket.UnitTests.csproj

 Test Coverage Impact

Initial Coverage: 4.2%

Final Coverage: 46.6%

Coverage Change: +42.3pp

 Quality Indicators

Pass/Fail Summary: ☒ All tests build and pass

 Insights & Recommendations

디버거



GitHub Copilot Chat

Analyze the value of the expression 'choice'

Using `@ #Debugger`, explain the value of the expression
choice

 Debugger

> 2 references

I'll help you analyze the value of the **choice** variable. Let me start by understanding the context of your program.

> ✓ Retrieved 'ConsoleApp3.UI.ConsoleUI'

Now let me check the switch statement in the Main method to see how this choice is being handled:

✓ Read Program.cs

Perfect! Now I have all the context I need to analyze the **choice** variable.

Analysis

Value: "1" (string)

What it represents: The user's menu selection indicating they want to **Add a new Todo** item to the application.

Why you're getting this value:

The program displayed a menu with 8 options via **ui.DisplayMenu()**. The user typed "1" at the console prompt and pressed Enter. **Console.ReadLine()** captured this input as the string "1". The debugger has paused at the switch

FileEditViewGitProjectBuildDebugTestToolsExtensionsWindowHelpSearchTimeZoneHelperWebApp

Process: [40208] TimeZoneHelpLifecycle EventsThread:Stack Frame:

TimeZoneservice.csTimeZoneHelperWebApp.WebTimeZoneHelperWebApp.Web.Services.TimeZoneServiceGetTimeZoneDisplayInfo(DateTime localDateTime, string localT

17// Convert local time to UTC first
18var utcTime = TimeZoneInfo.ConvertTimeToUtc(localDateTime, localTimeZone);
19
20var displayInfoList = new List<TimeZoneDisplayInfo>();
21
22foreach (var tz in _timeZones)
23{
24var tzInfo = TimeZoneInfo.FindSystemTimeZoneById(tz.Id);
25var convertedTime = TimeZoneInfo.ConvertTimeFromUtc(utcTime, tzInfo);
26
27displayInfoList.Add(new TimeZoneDisplayInfo
28{
29DisplayName = tz.Name,
30TimeZoneId = tz.Id,
31LocalTime = convertedTime,
32IsWorkingHours = IsWithinWorkingHours(convertedTime)
33});
34}
35
36return displayInfoList.ToArray();
37
38
391 reference, 0 changes, 0 authors, 0 changes
40public bool IsWithinWorkingHours(DateTime dateTime)
41{
42var hour = dateTime.Hour;
43return hour >= 9 && hour < 17; // 9am to 5pm (17:00)
44}
450 references, 0 changes, 0 authors, 0 changes
46public string[] GetAvailableTimeZones()
47{
48// Return commonly used time zones
49return new[]
50{
51"Pacific Standard Time",
52"Eastern Standard Time",
53"Central European Standard Time",
54

Diagnostics Tools
Diagnostics session: 10 seconds
Add counter graphs by checking counters from counter options
Events
Process Memory (MB)
CPU (% of all processors)

SummaryEventsCountersMemory Usage
Events
Show Events (0 of 0)
Application Insights Events (0 of 0)
.NET Counters
View Performance Counters
Memory Usage
Take Snapshot
CPU Usage

GitHub Copilot Chat
New Thread
Copilot Chat
I might make mistakes, so check for accuracy.
Suggest how I can optimize my code
Write unit tests for all the methods in #file
How can I improve #method? Suggest code for any improvements
Use # for references, / for commands, and @ for extensions.
/help what can you do?
+ Reference
AskClaude Sonnet 4.5

Autos
Search (Ctrl+E)
NameValueType
AutosLocalsWatch 1

Developer PowerShell 7
+ Developer PowerShell 7
tibility purposes. If you want to re-enable it, run 'Import-Module PSReadLine'.
** Visual Studio 2026 Developer PowerShell v18.0.0-insiders
** Copyright (c) 2025 Microsoft Corporation
PS C:\Users\rachelkang\source\repos\TimeZoneHelperWebApp>

100 %No issues foundLn: 36, Ch: 9SPCCRLFWindows 1252

1 inch of rain
Thursday

Search

0 / 06mainTimeZoneHelperWebApp

프로파일러



GitHub Copilot Chat

@Profiler Suggest how I can optimize my code

@Profiler Suggest how I can optimize my code

> 1 reference

Checkpoint created. [Restore](#)

Profiler

> 17 references

I'll help you optimize your code. Let me first understand the complete structure of your Todo application and then create benchmarks to identify performance bottlenecks.

> 4 tasks done

Now I'll create a plan to benchmark and optimize your code:

Plan Created

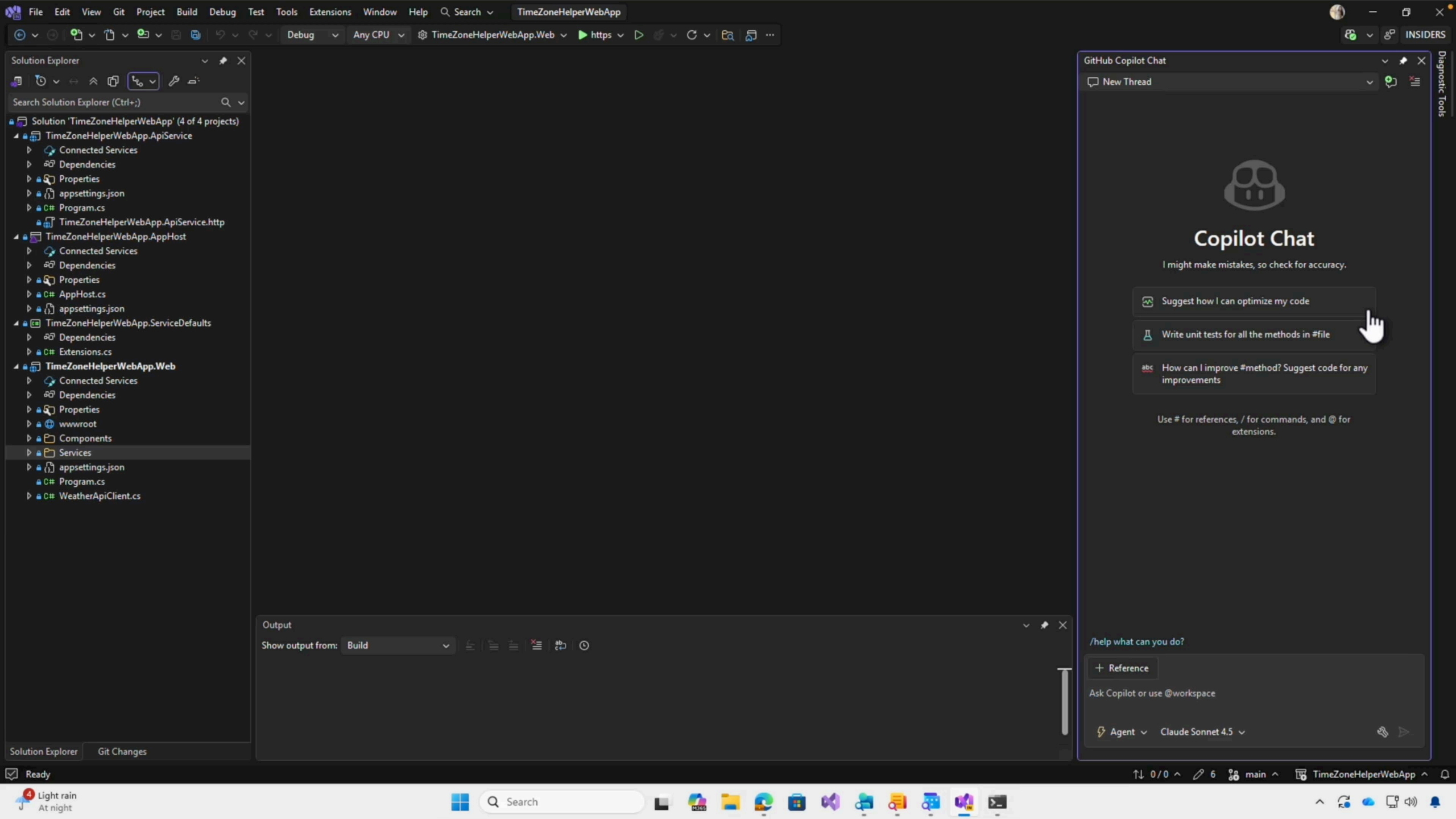
[Open](#)

I'll begin by:

1. Creating a benchmark project to measure the performance of your TodoService and TodoRepository
2. Running CPU benchmarks to establish baseline performance
3. Analyzing the results to identify bottlenecks
4. Optimizing the code based on the benchmark data
5. Re-running benchmarks to verify improvements

Let me start by creating benchmarks for the key operations in your application.

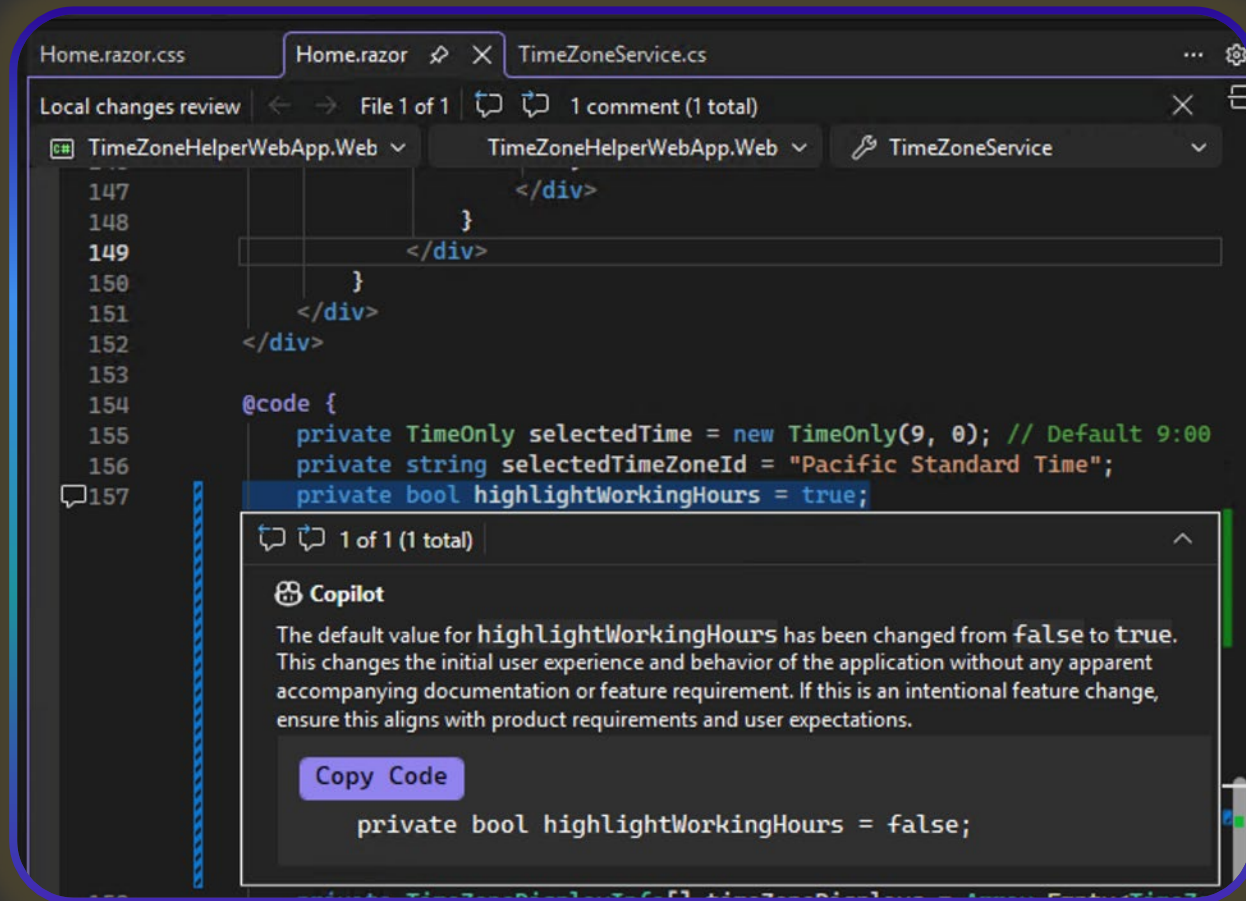
Step 1: Create benchmark project for



코드 리뷰



리뷰



FileEditViewGitProjectBuildDebugTestExtensionsWindowHelpSearchTimeZoneHelperWebApp

DebugAny CPUTimeZoneHelperWebApp.Webhttps

TimeZoneHelperWebApp

Git Changes - TimeZoneHelperWebApp

main

Enter a message <Required>

Commit AllAmend

Related Items

Changes (2)

C:\Users\rachelkang\source\repos\Ti...
TimeZoneHelperWebApp.Web\Co...
Home.razor
Home.razor.css

Stashes

Home.razorHome.razor.cssplan-c7e4620...25d55c6a7.mdProgram.csExtensions.csTimeZoneService.cs

TimeZoneHelperWebApp.Web

TimeZoneHelperWebApp.Web.Components.Pages.HTimeZoneService

```
86
87
88     if (highlightWorkingHours && timeZoneDisplays.Length > 0)
89     {
90         <div class="working-hours-summary">
91             @{
92                 var workingLocations = timeZoneDisplays.Where(tz => tz.IsWorkingHours).Select(tz => tz.DisplayName).ToList();
93                 var offHoursLocations = timeZoneDisplays.Where(tz => !tz.IsWorkingHours).Select(tz => tz.DisplayName).ToList();
94             }
95
96             if (workingLocations.Any() && offHoursLocations.Any())
97             {
98                 <p>
99                     <strong>Coworkers in @string.Join(", ", workingLocations)</strong> are still online.
100                     It's currently outside working hours for coworkers in <strong>@string.Join(", ", offHoursLocations)</strong>.
101                 </p>
102             }
103             else if (workingLocations.Any())
104             {
105                 <p>
106                     <strong>Coworkers in @string.Join(", ", workingLocations)</strong> are online.
107                 </p>
108             }
109             else if (offHoursLocations.Any())
110             {
111                 <p>
112                     It's currently outside working hours for all locations.
113                 </p>
114             }
115         </div>
116
117         <div class="timezone-grid">
118             @foreach (var tzInfo in timeZoneDisplays)
119             {
120                 <div class="timezone-card @GetCardClass(tzInfo)">
121                     <div class="timezone-header">
122                         <h3>@tzInfo.DisplayName</h3>
123                         <div class="timezone-header-actions">
124                             <span class="timezone-offset">@tzInfo.OffsetDislay</span>
125                         </div>
126                     </div>
127                 </div>
128             }
129         </div>
130     }
131 }
132
```

Ln: 98, Ch: 33 SPC CRLF UTF-8 with BOM

Output

Show output from: Debug

TimeZoneHelperWebApp.Web.exe (CoreCLR: clrhost): Loaded 'C:\Program Files\dotnet\shared\Microsoft.NETCore.App\10.0.0-rc.2.25502.107\System.IO.Compression'.
Microsoft.AspNetCore.Mvc.ViewFeatures.BrowserRefreshMiddleware: Debug: Response markup is scheduled to include browser refresh script injection.
Microsoft.AspNetCore.Mvc.ViewFeatures.BrowserRefreshMiddleware: Debug: Response markup was updated to include browser refresh script injection.
Microsoft.AspNetCore.Mvc.ViewFeatures.BrowserRefreshMiddleware: Debug: Script injected: /_framework/aspnetcore-browser-refresh.js
The program 'C:\Users\rachelkang\source\repos\TimeZoneHelperWebApp\bin\Debug\TimeZoneHelperWebApp.Web.exe' has exited with code 4294967295 (0xffffffff).

Solution Explorer

Git Changes

Error List

Output

GitHub Copilot Chat

Make the checkbox checked on by default

Home.razor.css TimeZoneHelperWebAp... -0 +21

Now let me validate the changes:

Read errors in TimeZoneHelperWebApp.Web/Components/Pages...

Perfect! I've successfully implemented the feature. Here's what I added:

Changes Made:

1. Added Working Hours Summary - When the "Highlight Working Hours" checkbox is checked (which is now on by default), a summary blurb appears that:

Shows which coworker locations are still online (in working hours)

Shows which locations are outside working hours

Displays a friendly message like: "Coworkers in Pacific (Redmond, WA), Eastern (New York) are still online. It's currently outside working hours for coworkers in Central Europe (Spain), Australia Eastern (Sydney)."

2. Styled the Summary Section - Added CSS styling with:

A warm yellow/orange gradient background

Clear border and shadow for visibility

Bold text to highlight location names

Positioned between the controls and the time zone cards

The summary dynamically updates whenever you change the time or time zone, and only appears when the checkbox is checked and there are time zones to display.

Total changes (2)

Keep Undo

Home.razor.css

Home.razor

Reference

Ask Copilot or use @workspace

Agent Claude Sonnet 4.5

Ready

48°F Cloudy

Search

Taskbar icons

System tray

Azure MCP Server



GitHub Copilot Chat

Deploy to Azure

Deploy to Azure
> 1 reference

GitHub Copilot
> 1 reference

I'll help you deploy this ASP.NET Core web application to Azure. Let me start by gathering some information about your project and then guide you through the deployment process.

✓ Read 1 project

✓ Ran verify_azure_publish_compatibility

Running Azure Best Practices

Run Azure Best Practices

Tool arguments:

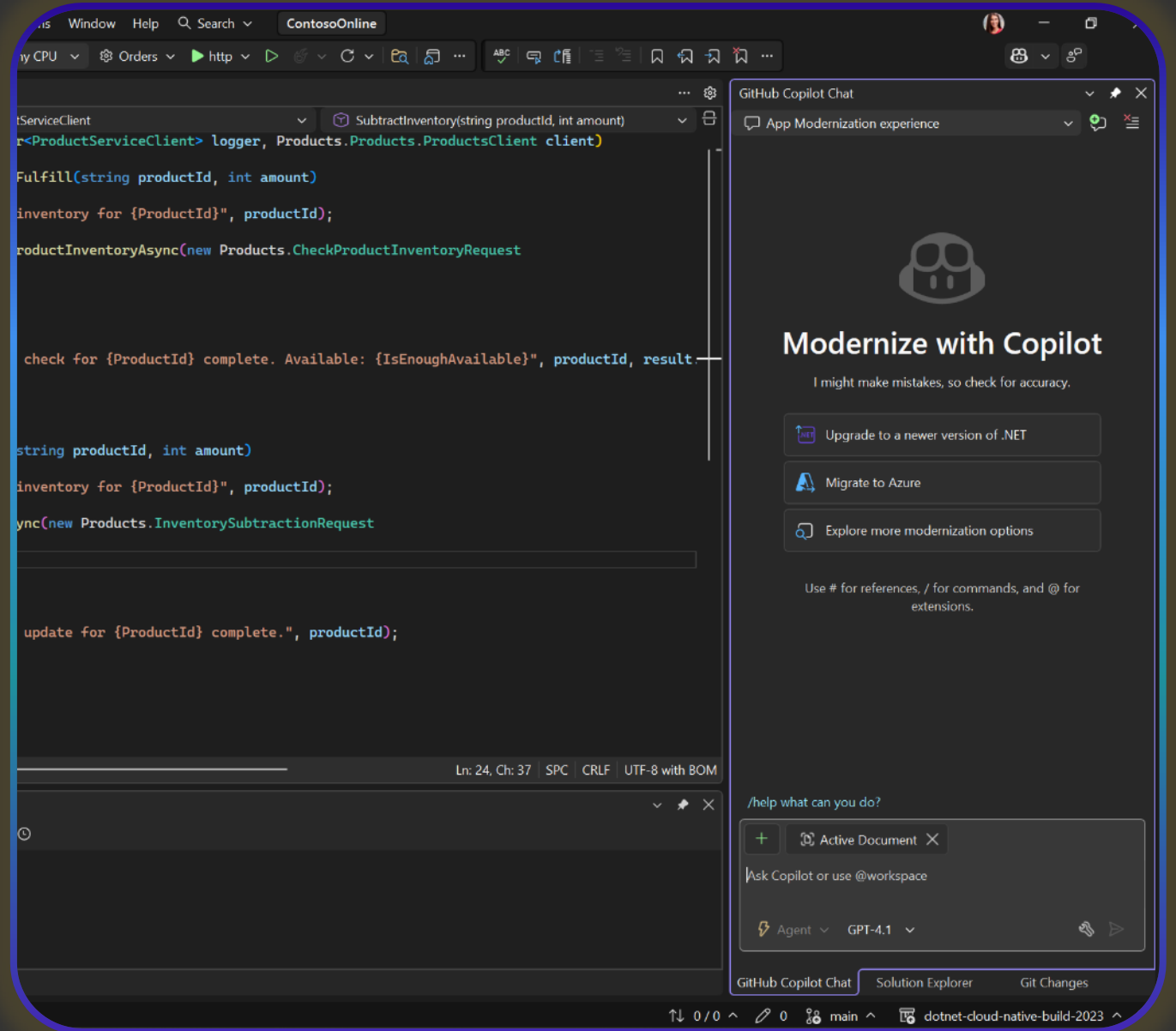
Name	Value
intent	Deploy ASP.NET Core web application to Azure

This tool is from 'Azure MCP Server' (MCP Server). Note that MCP servers or malicious conversation content may attempt to misuse Visual Studio through tools. Please carefully review any requested actions.

Confirm

Deny

현대화



GitHub Copilot Dev Days

Demo

Visual Studio + GitHub Copilot

감사합니다